

Question 1 (10 Marks)

Marks

- a) Simplify $\frac{a^2 - 9}{(a+3)^2}$ 1
- b) Expand and simplify $(a+b)^2 - (a-b)^2 + (a-b)(a+b)$ 2
- c) Factorize completely $x^4 - 16$ 2
- d) Simplify $\frac{x^2 + 5x + 6}{x^2 - 4} \div \frac{x^2 - 1}{x^2 - x - 2}$ 3
- e) Factorize completely $8x^3 - 27y^3$ 2

Question 2 (10 Marks) *Start a new page*

- a) Solve $x^2 + 6x - 1 = 0$ by completing the square, leaving your answer in simplest surd form. 2
- b) Express $x = 0.\dot{3}\dot{6}$ as a rational number. 2
- c) The weekly wages of five carpenters and three apprentices amount to \$1880 and the wages of three carpenters and five apprentices amounts to \$1640. By constructing a pair of simultaneous equations, find the weekly wages of both a carpenter and an apprentice. 3
- d) Find the values of a and b if $a + b\sqrt{7} = \frac{1}{7 + \sqrt{7}}$ 3

Question 3 (10 Marks) *Start a new page*

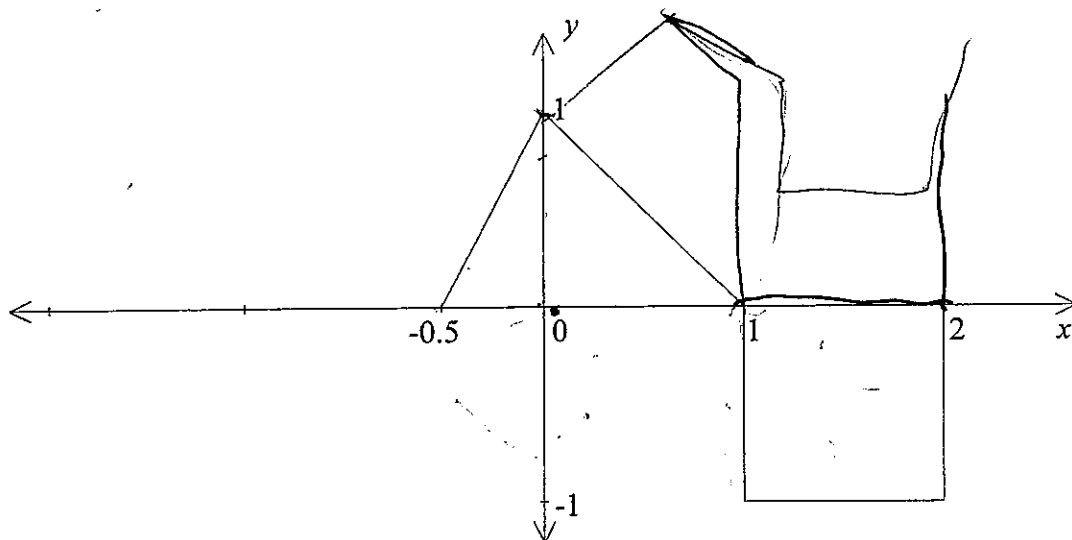
- a) Let $f(x) = 2x + 5$
- i) Find $g(x)$, the inverse of $f(x)$. 1
- ii) Show that $f(g(x)) = g(f(x))$ 2
- b) Let $y = 9x - x^3$
- i) Sketch the graph of this function clearly showing all intercepts. 3
- ii) From your graph, write down the region for which $9x - x^3 < 0$ 1
- c) Sketch $x + |y + 1| = 1$ indicating all important points. 3

Question 4 (10 Marks) *Start a new page*

Marks

- a) Graph $f\left(x + \frac{1}{2}\right) + 1$, if $f(x)$ has the following graph.

2



- b) Sketch $y = \sqrt{3-x}$

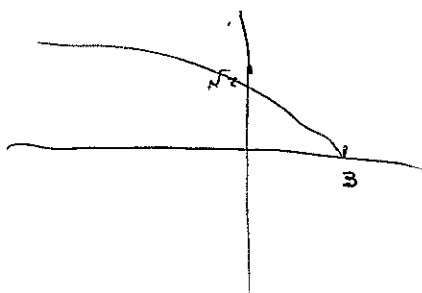
2

- c) Solve the inequation: $\frac{5}{x-4} \geq 1$.

3

- d) Prove whether the function $f(x) = e^x + \frac{1}{e^x}$ is odd, even or ~~neither~~.

3



END OF PAPER

11 EXT 1

Q1 a) $\frac{a^2-9}{(a+3)^2} = \frac{(a-3)(a+3)}{(a+3)^2} = \frac{a-3}{a+3}$ (1M) (OK)

b) $(a+b)^2 - (a-b)^2 + (a-b)(a+b)$
 $= a^2 + 2ab + b^2 - (a^2 - 2ab + b^2) + a^2 - b^2$ (1M)
 $= \cancel{a^2} + 2ab + \cancel{b^2} - \cancel{a^2} + 2ab - \cancel{b^2} + a^2 - b^2$
 $= a^2 + 4ab - b^2$ (1M)

c) $x^4 - 16 = (x^2 - 4)(x^2 + 4)$ (1M) $= (x-2)(x+2)(x^2 + 4)$ (1M)

d) $\frac{x^2 + 5x + 6}{x^2 - 4} \div \frac{x^2 - 1}{x^2 - x - 2} = \frac{(x+2)(x+3)}{(x+2)(x-2)} \times \frac{(x+1)(x-2)}{(x-1)(2+1)}$ (1M Factorising) (1M answer correct)
 $= \frac{x+3}{x-1}$ (1M inverting)

e) $8x^3 - 27y^3 = (2x - 3y)(4x^2 + 6xy + 9y^2)$ (1M correct terms) (1M correct signs)

Q2 a) $x^2 + 6x - 1 = 0$

$x^2 + 6x + 9 = 1 + 9$ (1M)

$(x+3)^2 = 10$

$x+3 = \pm\sqrt{10}$

$x = -3 + \sqrt{10}, -3 - \sqrt{10}$ (1M)

b) let $x = 0.363636 \dots$ (1M)

$\therefore 100x = 36.363636 \dots$

$\therefore 99x = 36$

$x = \frac{36}{99} = \frac{4}{11}$ (1M)

c) wages of carpenters = c wages of apprentices = a

(1M) $\begin{cases} 5c + 3a = 1880 \times 5 \\ 3c + 5a = 1640 \times 3 \end{cases}$

(1M) $\begin{cases} 25c + 15a = 9400 \\ 9c + 15a = 4920 \end{cases}$
 $16c = 4480$

$c = 280 \therefore a = 160$ (1M)

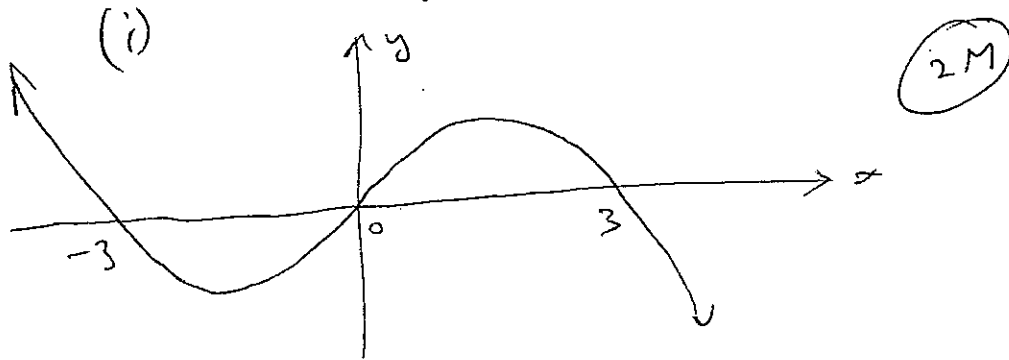
d) $a + b\sqrt{7} = \frac{1}{7+\sqrt{7}} \times \frac{7-\sqrt{7}}{7-\sqrt{7}} = \frac{7-\sqrt{7}}{42}$ (1M)
 $\therefore a = \frac{1}{6} \quad b = -\frac{1}{42}$ (1M)

3 a) $f(x) = 2x + 5$
 $y = 2x + 5 \quad \therefore x = \frac{y-5}{2} \quad \therefore y = \frac{x-5}{2}$

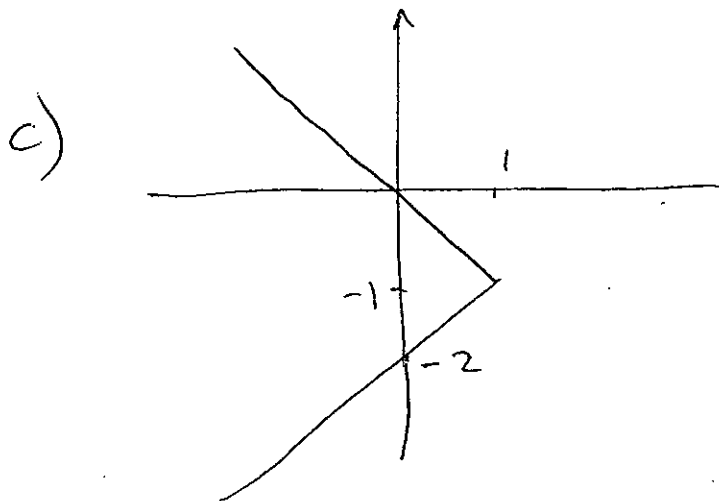
(i) $\therefore g(x) = \frac{x-5}{2}$ (1M)

(ii) $f(g(x)) = f\left(\frac{x-5}{2}\right) = 2\left(\frac{x-5}{2}\right) + 5 = x$ (1M)
 $g(f(x)) = g(2x+5) = \frac{2x+5-5}{2} = x$ (1M)

b) $y = 9x - x^3$ $-x^3 \downarrow$
 $= x(9 - x^2) = x(3-x)(3+x)$ (1M)
 x intercepts are $0, 3, -3$
 y intercept is $y = 0$



(ii) $9x - x^3 < 0$
 $\therefore x > 3 \text{ or } -3 < x < 0$ (1M)



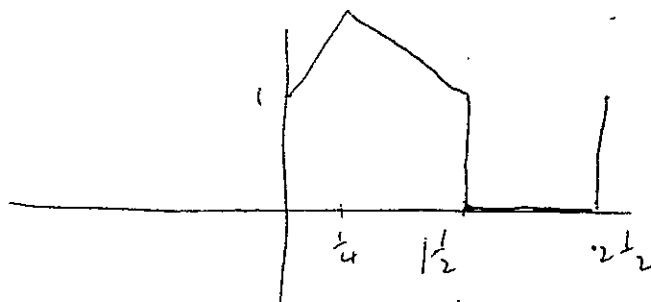
$x + |y + 1| = 1$

(1M) vertex coords correct

(1M) correct y intercepts

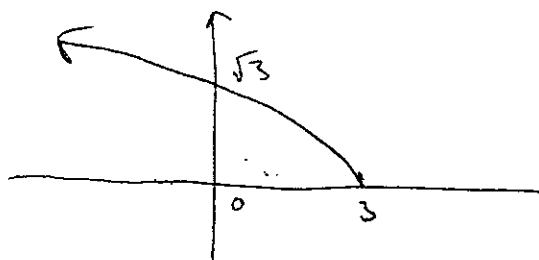
(1M) correct sketch

Q4 a)



1 each
for vert/horiz
shifts

b)



$$y = \sqrt{3-x}$$

$$3-x \geq 0$$

$$x \leq 3$$

c)

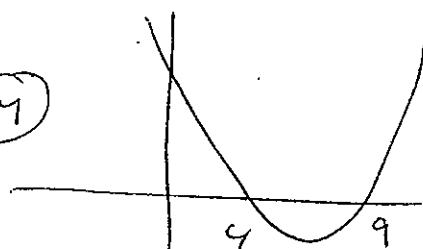
$$\frac{5}{x-4} (x-4)^2 \geq (x-4)^2 \leftarrow (1M) x \neq 4$$

$$(x-4)(x-4-5) \leq 0$$

$$(x-4)(x-9) \leq 0 \quad (1M)$$

$$D: 4 < x \leq 9$$

(I suggest we don't do too harsh about $x \leq 4$ at this stage. $\uparrow$)



d)

$$f(x) = e^x + \frac{1}{e^x}$$

$$f(-x) = e^{-x} + \frac{1}{e^{-x}}$$

$$= \frac{1}{e^x} + e^x$$

$$= f(x)$$

$\therefore f(x)$ is even

(1M)

(1M)

(1M)